

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (Civil) (Credit System) Degree Examinations**

January 2023

20CV503 - GEOTECHNICAL ENGINEERING - I

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*
1. a)	Derive an expression $V_d = \frac{(1-na)G.V_w}{1+\omega G}$	6	L*1	1
b)	With the help of phase diagram, define the terms water content, porosity and degree of saturation.	4	L1	1
c)	Draw a neat sketch of plasticity chart and explain the same for classifying the fine grained soil as per Indian standards.	10	L2	2
2. a)	A soil sample has a porosity of 40 percent. The specific gravity of solids is 2.7. Calculate i) Dry unit weight of sand, ii) Unit weight of sand, if $S = 0.5$ iii) Unit weight if the sand is completely saturated.	10	L3	1
b)	Explain the types of soil structures with neat sketch.	6	L2	2
c)	Write a note on base exchange capacity.	4	L2	2
3. a)	List the index properties of soil and explain determination of liquid limit using Casagrande apparatus.	6	L2	1
b)	Define the following terms: i) Shrinkage limit ii) Consistency Index	4	L2	1
c)	i) How silt is distinguished from clay in the field ii) Write a note on clay mineral montmorillonite	6 4	L2 L2	2 2
Unit – II				
4. a)	Distinguish between standard proctor and modified proctor test.	4	L3	3
b)	Briefly explain the factors affecting permeability.	6	L2	3
c)	Explain Casagrande method of determination of pre-consolidation pressure and define co-efficient of consolidation.	10	L3	4
5. a)	The following data was obtained from a standard laboratory proctor computer tests. Water content (%) - 8.5, 12.2, 13.75, 15.5, 18.2, 20.2 Weight of wet sample (kg) = 1.8, 1.94, 2.00, 2.05, 2.03, 1.98. The specific gravity of soil was 2.77, the container is 944 cm ³ in volume and 1.978 kg in weight. i) Plot the compaction curve and determine OMC ii) Plot 80% and 100% saturation lines	10	L3	3
b)	Discuss the assumptions and limitations of Terzaghi's one dimensional consolidation theory.	6	L2	4
c)	Differentiate between consolidation and compaction.	4	L3	4
6. a)	Briefly explain the factors affecting compaction.	6	L2	3
b)	Write a note on field compaction.	4	L2	3
c)	A soil sample 20mm thick takes 20 minutes to reach 20% consolidation. Find the time taken for a clay layer 6m thick to reach 40% consolidation. Assume double drainage in both cases.	10	L3	4
Unit – III				
7. a)	Explain the Mohr Coulomb failure theory and shear strength parameters of soil.	10	L2	5
b)	Classify the different methods to determine the shear strength of soil and explain any one method with neat sketch.	10	L2	5
8. a)	Discuss the factors affecting the shear strength of soil.	8	L2	5
b)	Explain the sensitivity and thixotropy of soil.	6	L2	5
c)	An unconfined compression test was conducted on as undistributed sample of clay. The sample had a diameter of 37.5mm and was 80mm long. The load at failure measured by the proving ring was 28 n and the axial deformation of the sample at failure was 13mm. Determine the unconfined compressive strength and undrained shear strength of the clay.	6	L4	5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
